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Q1. What is the 3dB concept in frequency response? Why do we take 3 dB less from the maximum gain?

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A1. Expressions for voltage and current gains of a common-emitter amplifier are complex-valued on account of capacitive impedances. The gain falls at lower frequencies where the reactances of the large coupling and emitter bypass capacitors

ing amplifiers.

The frequency response of the amplifier is plotted on the basis of the mid-frequency range, lower half-power frequency (f_L) and upper half-power frequency (f_H) as shown in Fig. 1. The gain magnitude is essentially constant over the mid-frequency range. By definition, the power gain magnitude at f_L and f_H is half of the value at the mid-frequency range. Also, at these frequencies, the voltage gain magnitude is $1/\sqrt{2}$ or 0.707 of its mid-frequency value.

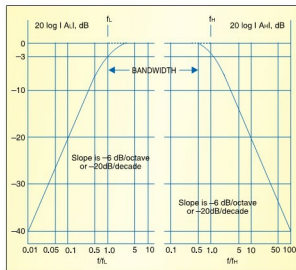


Fig. 1: A semi-log plot of the amplitude-frequency response of an RC-coupled amplifier

are large enough. Similarly, the gain falls at high frequencies on account of reactance of inter-electrode capacitors. The gain of an amplifier is therefore related to its frequency-handling capabilities.

A plot showing the magnitude of the voltage gain of an amplifier as a function of frequency will give you an idea how well the various frequency components in a signal are amplified. The gain magnitude is essentially related to frequencies in the base band, which can vary from 20 to 20,000 Hz for a hi-fi stereo system, from 300 to 3000 Hz for transmission of speech on telephone lines, and from 30 Hz to 4 MHz in the case of a broadband video amplifier. Besides gain and distortion design, the bandwidth becomes an important parameter while design-

ing the power-level difference in decibels at f_L and mid-frequency value is:

$$10 \log (P_L/P_{mid}) = 10 \log (1/2) = 10(-0.3) \text{ dB} = -3 \text{ dB}$$

That is, the power level of f_L is 3 dB down compared to the power level at mid-frequency range (0 dB). Similarly, the power level of f_H is 3 dB down compared to the power gain at the mid-frequency range.

f_L and f_H are sometimes called 'half power points.' The frequency range from f_H to f_L is called 'bandwidth' of the amplifier. A signal, with Fourier components that lie within f_L to f_H and have appreciable amplitude, will pass through the stage without excessive distortion.

Q2. What is sampling and how is it used to convert digital